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Modeling the environmental fate of bracken toxin ptaquiloside: Production, release and transport in the rhizosphere

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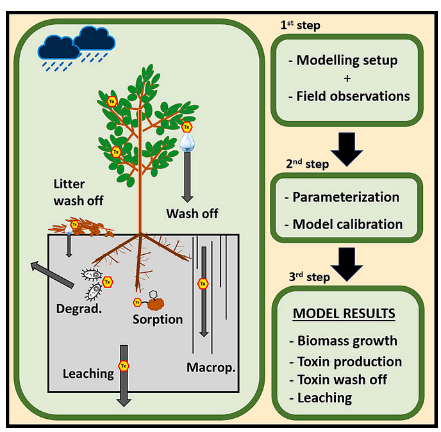
HIGHLIGHTS

- New DAISY model functions for simulating ptaquiloside (PTA) generation, wash off and degradation in soil.
- Bracken biomass growth, PTA generation and wash off are well described by extended model.
- PTA bracken-to-soil transfer and leaching in form of pulses linked to precipitation events.
- Macropore transport contributes to 72 % of the total mass of PTA leached.

GRAPHICAL ABSTRACT



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ABSTRACT

Plants produce a diverse array of toxic compounds which may be released by precipitation, explaining their wide occurrence in surrounding soil and water. This study presents the first mechanistic model for describing the generation and environmental fate of a natural toxin, i.e. ptaquiloside (PTA), a carcinogenic phytotoxin produced by bracken fern (*Pteridium aquilinum* L. Kuhn). The newly adapted DAISY model was calibrated based on two-year monitoring performed in the period 2018–2019 in a Danish bracken population located in a forest glade. Several functions related to the fate of PTA were calibrated, covering processes from toxin generation in the canopy, wash off by precipitation and degradation in the soil. Model results show a good description of observed bracken biomass and PTA contents, supporting the assumption that toxin production can be explained by the production of new biomass. Model results show that only 4.4 % of the PTA produced in bracken is washed off by precipitation, from both canopy and litter. Model simulations showed that PTA degrades rapidly once in the soil,

Abbreviations: PTA, ptaquiloside; PSM, plant secondary metabolites; SHP, soil hydraulic properties; VGM, Van-Genuchten-Mualem model; LAI, leaf area index; dw, dry weight.

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especially during summer due to the high soil temperatures. Leaching takes place in form of pulses directly connected to precipitation events, with maximum simulated concentrations up to 4.39 μg L^{-1} at 50 cm depth. Macropore transport is mainly responsible for the events with the highest PTA concentrations, contributing to 72 % of the total mass of PTA leached. Based on the results, we identify areas with high density of bracken, high precipitation during the summer and soils characterized by fast transport, as the most vulnerable to surface and groundwater pollution by phytotoxins.

1. Introduction

Plants are known to produce a high diversity of plant secondary metabolites (PSM) that serve a wide array of functions for the plant such as protection against biotic or abiotic stressors, i.e. pests, herbivores or drought. Many PSMs are also involved in plant-microbe signaling and e.g. are linked with the development of mycorrhiza (Sharma and Verma, 2018; Akula and Ravishankar, 2011; Hartmann, 2004; Isah, 2019). Based on the toxicity to animals as well as humans, PSMs have been commonly referred to as phytotoxins in an environmental context (Bucheli, 2014; Hartmann et al., 2008). A high number of phytotoxins have been detected in soils (Hama and Strobel, 2019; Rasmussen et al., 2003a), surface water (Kisielius et al., 2020; Clauson-Kaas et al., 2016), and groundwater (Kisielius et al., 2020; Tung et al., 2020; Skrbic et al., 2021) originating from known toxin-producing plants and with concentrations up to milligrams per liter. Some relevant examples of phytotoxin chemical classes detected in the environment are alkaloids (*Lupinus* sp., *Petasites hybridus*, *Senecio* sp.), aristolochic acids (*Aristolochia* sp.) and terpenoids (*Pteridium* sp.) (Kisielius et al., 2020; Günthardt et al., 2020a; García-Jorgensen et al., 2021). The wide occurrence of plants producing toxic compounds, the frequent detection of phytotoxins in soils and water and their intrinsic mobility and toxicity, have led to an increasing attention on phytotoxins as potential environmental pollutants (Bucheli, 2014; Günthardt et al., 2018; Günthardt et al., 2020b).

The most relevant processes determining the fate of phytotoxins in the environment are toxin production in the canopy, release from the plant, sorption to soil components, degradation, transport in soils and consequently leaching to groundwater (Fig. 1) (García-Jorgensen et al., 2020). The mass of phytotoxin produced by plants is affected by both biotic (pests, herbivores) and abiotic (light, drought, nutrients) factors, hence being extremely variable both spatially and temporally (Isah, 2019; Khare et al., 2020; Zandalinas et al., 2017). Phytotoxins present in plant tissue, both living plant or litter, can be washed off by precipitation, exudated through roots or released during decay of plant materials (García-Jorgensen et al., 2021; Albuquerque et al., 2011; Jessing et al., 2013; Rial et al., 2018).

The release mechanism of phytotoxins from the plant are poorly understood, both qualitatively and quantitatively (García-Jorgensen et al., 2021; Albuquerque et al., 2011). In the case of release by precipitation wash off, the amount being washed off has been suggested to be a function of the toxin content in the plant, canopy structure and precipitation volume and intensity (García-Jorgensen et al., 2021). Phytotoxins can also be actively exudated from roots, as some of these substances have been found to serve different functions for the plants, such as plant protection or inducing the formation of mycorrhiza, between others (Selmar et al., 2019a; Selmar et al., 2019b). Moreover, the remaining content of phytotoxins in the plant materials after senescence can be released to the soil during decay (García-Jorgensen et al., 2021; Hama and Strobel, 2020).

Once in the soil, phytotoxins can be sorbed and degraded by several mechanisms such as chemical hydrolysis (Ayala-Luis et al., 2006; Jiang et al., 2019), microbial metabolism (García-Jorgensen et al., 2021; Jessing et al., 2009), and irreversible sorption. Phytotoxins are vastly diverse in terms of occurrence, mobility, and persistence. Hence, the overall environmental fate is expected to be highly variably between chemical classes.

One of the best-studied phytotoxins is ptaquiloside (PTA), a major illudane-type glycoside produced by bracken fern (*Pteridium aquilinum* L. Kuhn) (Rasmussen et al., 2003a; Mori et al., 1985; O'Connor et al., 2019). Bracken is one of the most abundant vascular plants in the world with presence in all continents except Antarctica (Page, 1976). The toxicity of this fern is mostly attributed to the presence of PTA, which has been categorized by the International Agency for research on Cancer (IARC) in group 2B, as possibly carcinogenic to humans (International Agency for Research on Cancer (IARC), 1998). PTA is highly hydrophilic and weakly sorbing, as implied by a log octanol-water partitioning coefficient of −0.63 (Rasmussen et al., 2005). Considering the abundance of bracken, the toxicity and mobility of PTA, this phytotoxin has been categorized as a potential aquatic micropollutant (Günthardt et al., 2018). PTA has been detected in soils, surface waters and groundwater in bracken infested areas at concentrations up to mg L^{-1} (Rasmussen et al., 2003a; Clauson-Kaas et al., 2016; García-Jorgensen et al., 2021).

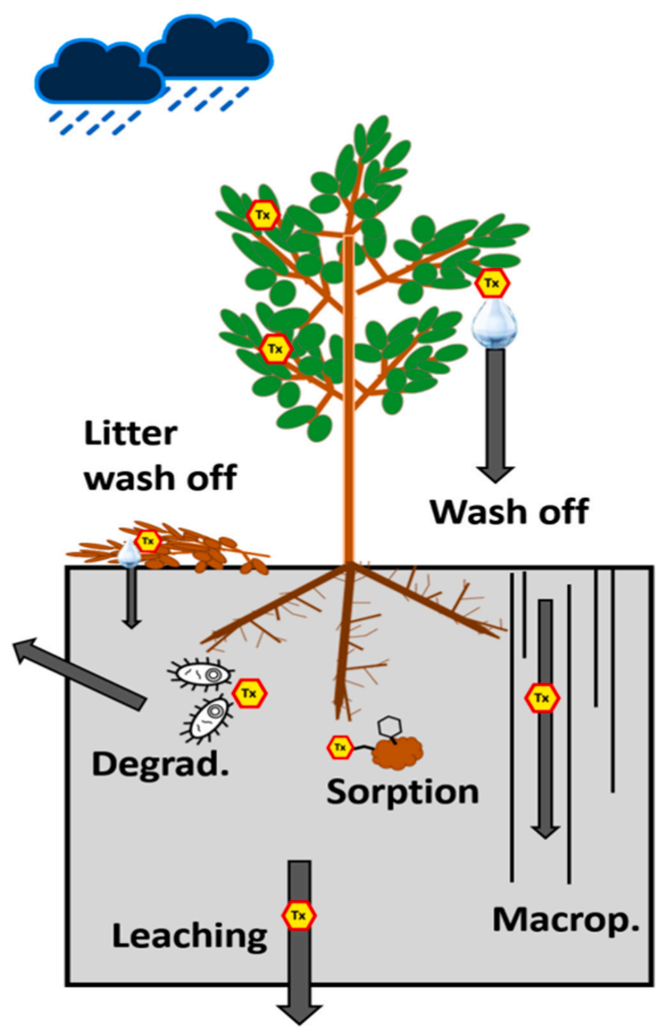

Fig. 1. Main processes contributing to the environmental fate of phytotoxins. These processes are wash off, degradation, sorption, macropore transport and leaching.

Previous studies proved that PTA can be effectively washed off by precipitation in high amounts, yielding concentrations in the drip off water up to 2530 μg L^{-1} (García-Jorgensen et al., 2021). Exudation of PTA from bracken rhizomes has not been detected in any previous study. Rhizomes act as storage organs where nutrients, and PTA, are stored during the winter period until the beginning of the growing season, where these nutrients are utilized to facilitate vigorous plant growth (Rasmussen et al., 2003b; Pakeman et al., 1994). However, there is no evidence of PTA exudation from bracken rhizomes, supporting the idea that leaf wash off is the predominant release mechanism of PTA from bracken (Clauson-Kaas et al., 2016; García-Jorgensen et al., 2021).

A novel conceptual approach for modeling the environmental fate of phytotoxins was presented in the study by García-Jorgensen et al. (2020). Based on previous literature, three new functions were implemented into the agroecosystem model DAISY to improve its applicability for the special case of bracken fern and PTA (Abrahamsen and Hansen, 2000). The new functions were: 1) the production of a natural toxin within the canopy as a function of the plant biomass, 2) the regulation of toxin wash off from different plant materials (e.g. canopy and litter), and 3) control of the toxin content at different plant development stages. The authors of this study performed a sensitivity analysis of the model to highlight how the implemented processes affected the overall max fluxes of PTA in the field. However, the lack of a coherent dataset with all relevant information needed for calibration made the predictions uncertain.

A recent study by García-Jorgensen et al. presented a two-year monitoring program (2018–2020) studying bracken growth, PTA content in the canopy, wash off during precipitation events and concentrations in soil solution (García-Jorgensen et al., 2021). The study presented the first attempt to conceptualize and quantify the dynamics of the different pools of PTA (e.g. plant materials, soil solution) and processes determining PTA fate (e.g. wash off, degradation, transport). Moreover, different environmental variables such as temperature, relative humidity and soil water content were monitored on-site. Peak concentrations observed in soil solution were three orders of magnitude above calculated maximum tolerable concentrations for PTA in drinking water (0.5–16 ng L^{-1}) (Rasmussen et al., 2005), hence raising concern of a plausible groundwater pollution. The long-term dataset with information regarding PTA, bracken and environmental variables represents an ideal dataset for testing the performance of a model previous adapted for phytotoxins (García-Jorgensen et al., 2020).

The aim of this study is to compare the adapted DAISY model for PTA and bracken (García-Jorgensen et al., 2020), against the 2-year monitoring dataset generated by García-Jorgensen et al. (2021). Special attention is given to the description of processes in the aboveground biomass such as phytotoxin production in the canopy and wash off by precipitation. The overall objective is to test the performance of the adapted DAISY model to describe PTA production from the top of the canopy to the bottom of the rhizosphere (50 cm depth).

2. Material and methods

2.1. Data and site description

The study by García-Jorgensen et al. presented a two-year monitoring of bracken and PTA (García-Jorgensen et al., 2021). The monitoring took place in Humleore (Zealand, Denmark), during the period ranging from May 2018 until March 2020. The climate at the study area is mild temperate, with significant rainfall distributed throughout the year and heavy showers concentrated in the summer period. An average temperature of 9.8 °C was recorded during the monitoring period. A total of 489 and 715 mm of precipitation were recorded for the year 2018 and 2019, respectively (Danish Meteorological Institute (DMI)). The soil profile in Humleore is classified as a sandy loam in the USDA texture classification and identified as an Oxyaquic Hapludalf in Soil Taxonomy (García-Jorgensen et al., 2021; USDA N, 2010; Ditzler et al., 2017). The soil has developed on morainic till and has an increasing content of clay and silt with depth.

Two monitoring plots, each of 25 m^2, were set in the center of a forest glade with bracken present as dominant species. In the glade, bracken appears as the dominant species with a dense canopy coverage surpassing two meters of height at its peak during summer (García-Jorgensen et al., 2021). In the two plots, bracken biomass and PTA concentration were monitored during the 2018 and 2019 growing seasons, with a frequency of sampling ranging from 2 to 3 weeks depending on the time of the year. To estimate the release of PTA by precipitation, water throughfall (i.e., water spilling off the canopy) was collected under the bracken canopy during four precipitation events. Concentration of PTA in the soil solution at 50 cm depth was monitored with suction cells during the period from March 2019 until March 2020. The volumetric water content in the soil profile was monitored with the same frequency as soil solution concentrations. The placement of the sensors is shown in Fig. 1 of the SI. For more details regarding the site, monitoring set up or data analysis, the reader is referred to García-Jorgensen et al. (2021).

2.2. The DAISY model

The DAISY model (1D, version 5.91) is a Soil-Plant-Atmosphere system model that simulates water, heat and solute balance, as well as plant production and yield under different management scenarios (Abrahamsen and Hansen, 2000; Hansen, 2002). The model also includes modules for simulating nitrogen cycling and organic matter turnover in the soil (García-Jorgensen et al., 2020). An advantage of the DAISY model compared with other advection-dispersion models is a generic plant growth module, simulating hourly phenological development and biomass production, as well as the development of a one-dimensional canopy.

In the following section, we briefly introduce the most relevant processes for the environmental fate of phytotoxins as modelled by DAISY. These processes are related to water flow, biomass production and environmental fate at the pedon scale. For a more comprehensive description of the different DAISY modules for water flow (Holbak et al., 2021), solute transport (Mollerup et al., 2014; Hansen et al., 2012a) and biomass production (Petersen et al., 1995; Børgesen and Olesen, 2011), the reader is referred to the literature.

2.3. Relevant DAISY submodules

In DAISY, water flow in the soil matrix is described by the Richards' equation (Richards, 1931), in which the soil hydraulic properties (SHPs), among others, can be represented with the Van-Genuchten-Mualem model (VGM) (Van Genuchten, 1980). Solute transport in the soil matrix is described with the advection-dispersion equation and with a non-local approach for the macropores (Eq. (5), SI) (Holbak et al., 2021). For more details regarding equations describing water flow and solute transport in DAISY see Supplementary Information (SI, Section 1) or specific literature (Holbak et al., 2021, 2022; Hansen et al., 2012b).

In Daisy, time of emergence of the plant is controlled by the sowing date, provided by the user and the temperature sum (T_{sum}). Plant phenomenological development in the model is described by the state variable Development Stage (DS). This has a value of 0 for emergence, 1 for flowering and 2 for maturity. The increase in development stage over time is calculated based on a constant rate, DSrate 1, between DS 0 and 1, and DSrate 2 between DS 1 and 2, which are further scaled by a modifier function based on air temperature (Hansen et al., 2012b). The effect of senescence (DS > 2) is simulated by a programmed reduction in the amount of available nutrients for plant growth, leading to a slow decrease in plant biomass. Death of the plant is simulated by a cutting action, which date is controlled by the mean air temperature and development stage. During the cutting actions, all the plant is deposited onto the soil profile becoming litter.

The net production of biomass is calculated based on gross photosynthesis, partitioning of nutrients within the plant and respiration from the different plant organs (Hansen, 2002). The canopy is a dynamic component of the biomass and it is defined by the leaf area index (LAI) [L^2 L^{-2}], which indicates the area of leaves per soil surface area. LAI is an important variable controlling evapotranspiration, photosynthesis and wash off by precipitation (Hansen, 2002).

García-Jorgensen et al. developed the plant model in DAISY to simulate the generation of natural products (toxins) within the plant (García-Jorgensen et al., 2020). Toxin generation is defined by a linear function based on biomass:

$$\frac{dM_{MAX}}{dt} = a\,f(DS)\,\frac{dB_{CAN}}{dt} \tag{1}$$

where M_{MAX} [M L^{-2}] is the potential toxin content in the canopy per unit area; B_{CAN}, [M L^{-2}] is the aboveground biomass, calculated by DAISY's plant model, and a [−] is a constant. The constant a determines the relationship between the toxin content and biomass. $f(DS)$ scales a based on the plant's development stage DS, thus controlling toxin generation throughout the different phenological stages. The PTA generation model developed by García-Jorgensen et al. (2020) implemented a process for toxin recovery after precipitation events. In this study we assume that the recovery of PTA in the canopy is not an active process and hence, we set the recovery rate to zero. This decision was supported by the observation that toxin production in bracken was rather correlated with the generation of new biomass than with wash off (García-Jorgensen et al., 2021).

In DAISY, chemical compounds present in the plant materials can be released to the soil exclusively by precipitation wash off. The amount of toxin being washed off is a function of the water dripping off the canopy and is calculated by:

$$W(t) = M(t)\left(1 - e^{-f_{wc}\left(\frac{D(t)}{Ic\ LAI(t)}\right)}\right) \tag{2}$$

where W [M L^{-2}] is the amount of toxin washed off the canopy, M [M L^{-2}] is the toxin content at the time of precipitation; f_{wc}, the canopy wash off coefficient; D [L^3 L^{-2}], the volume of drip off water; Ic [L^3 L^{-2}], the interception capacity of the canopy. Values of f_{wc} range from 0 to 1 and determine the amount of toxin that is washed off the plant by mm of water spilling off the canopy. The wash off equation (Eq. (2)) is similar to the one recommended by the FOCUS guidelines for pesticides (Boesten et al., 2000), with the only difference being that the potential volume of water intercepted in the canopy is dynamic and a function of LAI. Wash off from litter is calculated in the same way (Eq. (2)), but the values for LAI and water capacity are defined with specific parameters describing the properties of the litter layer (Section 2.2 in SI).

Degradation in the soil matrix is calculated by first-order kinetics based on a reference degradation rate, which is further scaled by different abiotic factors. Thus, the actual degradation rate is calculated as:

$$\xi = -[k^{*}\, f_T(\mathrm{T})\, f_h(\mathrm{h})\ f_Z(\mathrm{Z})] \tag{3}$$

where k^* [T^{-1}] is the reference degradation rate at standard conditions; T [°C], temperature; h [L], soil water pressure head and Z [L], soil depth. The f_T, f_h, f_Z coefficients are modifier functions for temperature, pressure head and soil depth (García-Jorgensen et al., 2020). The effects of temperature and soil moisture on degradation were described following FOCUS recommendations (Boesten et al., 2000). In the present study, the effect of soil depth on degradation was obtained from experimental degradation data for different soil depths. A power function was used to extrapolate the values for the modifier at deeper soil horizons than the observed depths (SI, Table S3) (García-Jorgensen et al., 2021).

2.4. Input data, parameterization and calibration of the model

2.4.1. Weather input data

The time period of the simulation was from January 2017 until September 2020, where the first year (2017) was used for warming-up of the model. Weather data were obtained from the closest weather station from the Danish Meteorological Institute (DMI). The data include air temperature [°C], relative humidity [%], precipitation [mm] and wind speed [m s^{-1}] with values in a ten-minute frequency; these data were later transformed to daily values for the final model input, except from precipitation that was transformed into hourly values. Global radiation [W m^{-2}] was obtained from the weather station of University of Copenhagen in Taastrup, located 30 km east of the study site. Reference evapotranspiration was calculated from the meteorological data with the Penman-Monteith equation recommended by FAO (Allen et al., 1998). Daisy calculates potential crop evapotranspiration based on the simulated dynamic LAI and the default crop factors of 1.2 for a full plant cover and 0.6 for bare soil.

2.4.2. Soil parameterization

We simulated a soil profile of 100 cm depth with five distinct horizons following the description of Humleore soil by García-Jorgensen et al. (2021) (see SI). The soil hydraulic properties (SHPs) for each horizon were calculated using the pedotransfer function HYPRES (Wösten and Nemes, 2004). A low conducting layer at 50–70 cm depth was included in the soil description (B_2) as part of the B horizon, to represent the heavy clayey layer observed in the field. For this, we assumed the B2 horizon to be identical to the B horizon, but with 100 times lower conductivity. Moreover, two classes of macropores were defined in the soil profile (both 5 mm in diameter), one class with macropores ranging from 10 to 80 cm depth (M_{80}) and a second class ranging from 0 to 55 cm (M_{55}). The first class allows water and solute movement over the low hydraulic layer (B_2), while the second class allows for fast movement of water from the soil surface to below the rhizosphere. Including these two classes was necessary to describe the measured water content profiles and to reproduce the field water balance. The bottom boundary was defined with a constant groundwater table at 100 cm in the soil. The choice of a fixed groundwater table as the bottom boundary was supported by the observed water dynamics at 94 cm depth (García-Jorgensen et al., 2021), which indicated high water content at depth throughout the year. This was found to be both a simple approach and, at the same time, offered a good fit against field observations.

2.4.3. Calibration of the model

The model was calibrated manually in three iterative steps as shown in Fig. 2. First, the soil model was calibrated to represent water dynamics in Humleore for the year 2019. After the model simulated satisfactory results for water contents, we calibrated the plant model to reproduce similar bracken biomass to the observations in 2018–19. In the last step, parameters controlling the processes contributing to the fate of PTA in the field (e.g. production, wash off and degradation) were calibrated to plant and soil water observations for the year 2018–2019.

In the first step, the density of macropores (D) was calibrated against observed water content at 34 and 54 cm depth. These densities are D_{80} and D_{55} for the M_{80} and M_{55} classes, respectively. These two depths for water content were selected as they are the two depths where soil water content was monitored most intensively, as well as at the depth where soil solution was sampled (50 cm).

In the second step, the plant module in DAISY was calibrated by hand to represent the observed biomass in Humleore for the years 2018 and 2019 (García-Jorgensen et al., 2021), as well as achieving realistic LAI values for that biomass (Pitman, 1989). The focus of this study was to simulate PTA production in bracken and wash off, hence biomass and LAI were the only two critical plant variables for simulating PTA fate. We decided to focus on the calibration of the PTA content in 2019 growing season, as we only had soil solution concentration data for 2019

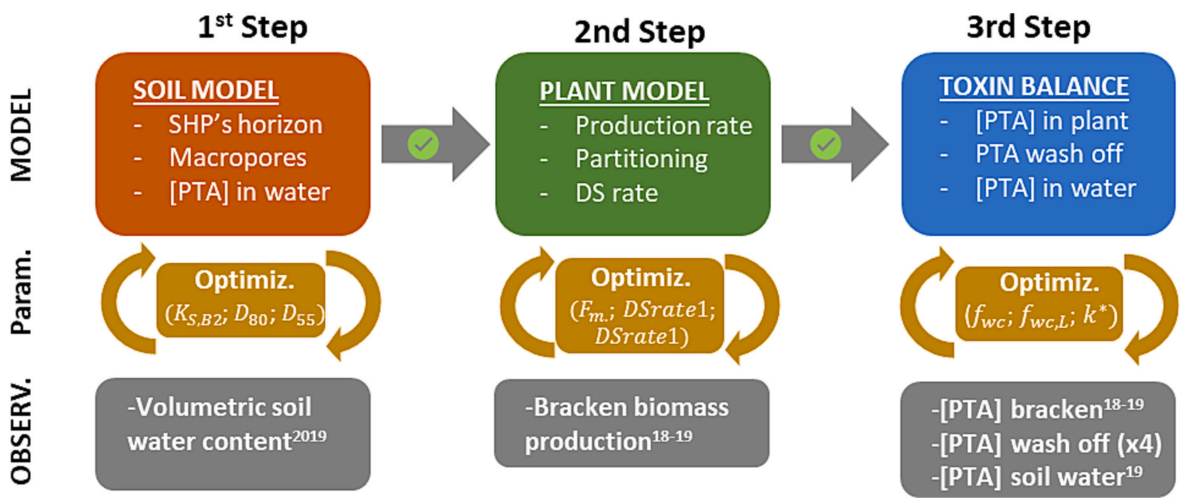

Fig. 2. Overview of the process for calibration of the model for soil, bracken and PTA. The top colored squares represent the different submodels. The yellow squares in the middle represent the parameters being calibrated. Grey squares represent observed data used for calibration of the model (year).

and we assumed that the PTA content in the 2018 growing season was strongly affected by the water stress in the plant.

As a starting point in step 2, we used the parameterization of the plant model by García-Jorgensen et al. (2020), which showed initially good results in terms of duration of the growing season for bracken in Humleore. Three parameters controlling biomass production (*Fm*) and development stage (*DSrate1* and *DSrate2*) were manually calibrated to represent the observations of bracken in the field. The maximum rooting depth was set equal to 45 cm, corresponding to field observations of bracken root density by the author (SI, Fig. S2). For more details regarding the plant model parameterization the reader is referred to the SI (2.1).

The third step was to calibrate all aspects related to PTA in the field, considering both plant and soil solution. The generation of PTA in the bracken canopy was calibrated to represent field observations (SI, Tables S2–3) of biomass PTA concentrations (García-Jorgensen et al., 2021). The function (f_{DS}, Eq. (1)) was used to simulate the constant decay of the toxin content in bracken biomass during later development stages (DS > 1.1), where we assume that bracken contains only 20 % of its potential maximum toxin content when the plant has reached maturity (DS 2). For more details regarding the values used for the parameterization of PTA production in the biomass the reader is referred to Section 2.2 in SI.

The parameters determining the fate of PTA in the soil, such as canopy wash off coefficient (f_{wc}), litter wash off coefficient ($f_{wc,L}$), and reference degradation rate (k^*), were fitted by comparing model results with measurements of the PTA wash off and PTA concentrations at 50 cm depth. Table 1 shows the fitted values for these five parameters are shown. Initially, f_{wc} and k^* were set to values measured in the monitoring program (García-Jorgensen et al., 2021) (Table 1).

Table 1
Fitted values for the different parameters calibrated in the model.

Step	Param.	Value	Units
1	$K_{S,B2}$	0.001	cm h^{-1}
1	D_{80}	50	m^{-2}
1	D_{55}	200	m^{-2}
3	f_{wc}	1×10^{-4}	–
3	$f_{wc,L}$	5×10^{-3}	–
3	k^*	0.7	h^{-1}

$K_{S,B2}$, saturated hydraulic conductivity of B_2 horizon; D_{80}, density of macropore class M_{80} D55; density of macropore class M_{55}; f_{wc}, canopy wash off coefficient; $f_{wc,L}$, litter wash off coefficient; k^*, base degradation rate.

However, as these values were uncertain, highly heterogeneous between events and resulted in a poor fit, the wash off coefficient from the canopy was calibrated against observations from the four different rain events that were monitored during 2018 and 2019. After calibrating the release from the plant, reference degradation rate was calibrated with the goal of simulating the PTA concentrations in the soil solution at 50 cm depth. To finalize the calibration process, the distribution densities for the two macropore classes (step 1) were fine tuned for optimization of the model fit against soil solution concentrations observed in 2019.

2.5. Extrapolation in time

We conducted a longer simulation with the final calibrated Daisy model to assess the environmental fate of PTA under different climatic conditions, i.e. weather variability between years. For that purpose, a single simulation for the period 2014–2019 was performed, where the first year was used for warming-up of the model. A total of five years (2015–2019) were considered to evaluate the contribution of different processes to the overall fate of PTA in the field, as well as to provide an estimate for the annual mass fluxes of PTA released to groundwater.

3. Results and discussion

3.1. Biomass production

DAISY describes very well the observed biomass production during the years 2018 and 2019 (Fig. 3) (García-Jorgensen et al., 2021). The simulated production of biomass is above 800 g m^{-2} during both years and reaching a maximum of 1136 g dry weight (dw) m^{-2} in 2019, in comparison with the observed maximum in the field of 1196 g dw m^{-2}. The differences in simulated production of biomass between years is attributed to water stress during the early development stages of bracken, especially during the exceptionally dry 2018 growing season. Model results show a significantly high number of days with water stress in the period ranging from May until August 2018, hence hampering biomass production when bracken growth is more vigorous. This effect of water stress on biomass production and development stage could not be explained with the current plant model parameterization, hence leading to slightly higher simulated maximum biomass values in 2018 compared to field observations.

García-Jorgensen et al. reported a heavy storm that tilted the bracken canopy at the end of July 2019, and biomass sampling was discontinued (García-Jorgensen et al., 2021). The disturbance caused by the collapsed canopy likely had a significant negative effect on bracken growth, for

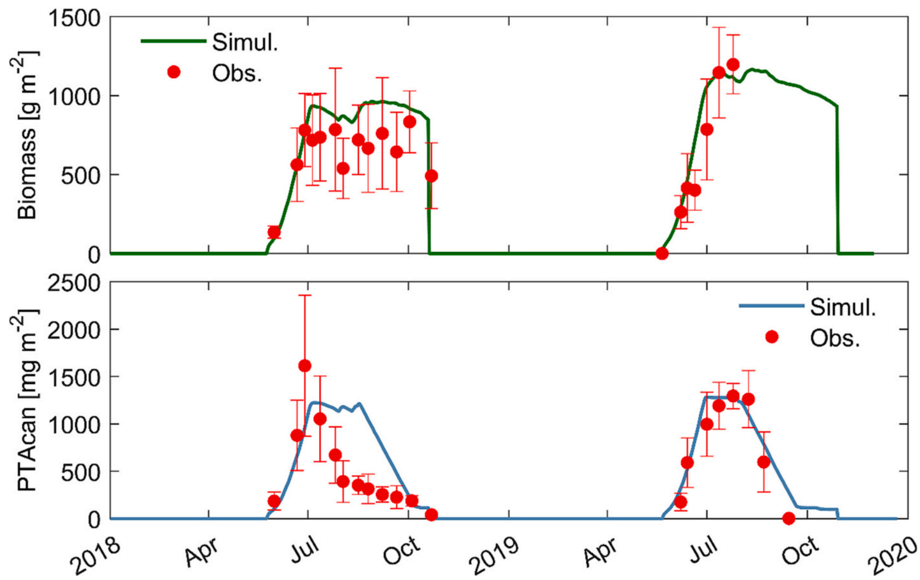

Fig. 3. Top: model fit of simulated biomass production (green line) for the 2018–2019 growing season compared with field observations (red points). Bottom: model fit of simulated ptaquiloside content in the canopy (PTAcan, blue line) for the 2018–2019 growing season compared with field observations (red points). Observation data points from García-Jorgensen et al. (2021), shown as average ± standard deviation (n = 6) for each sampling day.

instance by lower light availability in the more compacted canopy. It was not possible to continue the biomass monitoring after the storm, and hence there are no data to compare with the extended simulated biomass values for 2019. Disturbances of bracken like tilting of the canopy might have accelerated the senescence of bracken for 2019.

3.2. *Toxin production and release*

The simulated PTA contents in the plant agree well the dynamics observed in the field at different developmental stages (Fig. 3). The simulated production of PTA in the bracken canopy is above 1 g m^{-2} for both years, in agreement with the field observations (Fig. 3). Production of PTA is concentrated during the early development stages of bracken, and it ceases when there is no further net production of biomass (DS 1.25). From that point onwards, the PTA content in the canopy slowly decreases towards senescence (DS > 2).

A visual inspection of the model fits in Fig. 3 shows an overestimation of the maximum PTA contents simulated in 2018 compared to field observations. This was due to the incapacity of the model to simulate accurately the effect of environmental stressors, especially drought, on biomass growth and phenological development (DS), both being used to calculate PTA production in the model. The observed PTA content started decreasing earlier in the year 2018, but at a much lower rate compared to the observations in 2019, with significant amounts of PTA remaining in the canopy until late October. The different conditions observed in the field during the late growing season in 2019, e.g. the morphology of the decaying bracken materials or contact with soil materials, are expected to have a high influence on the concentration of PTA in bracken late in the season. The influence of canopy morphology on PTA dynamics were not considered in the model due to the lack of data, hence explaining the incapacity of the model to explain the PTA dynamics under different environmental conditions.

The simulated PTA washed off (Fig. 4) is a function of the amount of precipitation, bracken PTA content and LAI, as described previously. This correlation is apparent in the increased amounts of PTA being washed off per mm of precipitation through the season until bracken

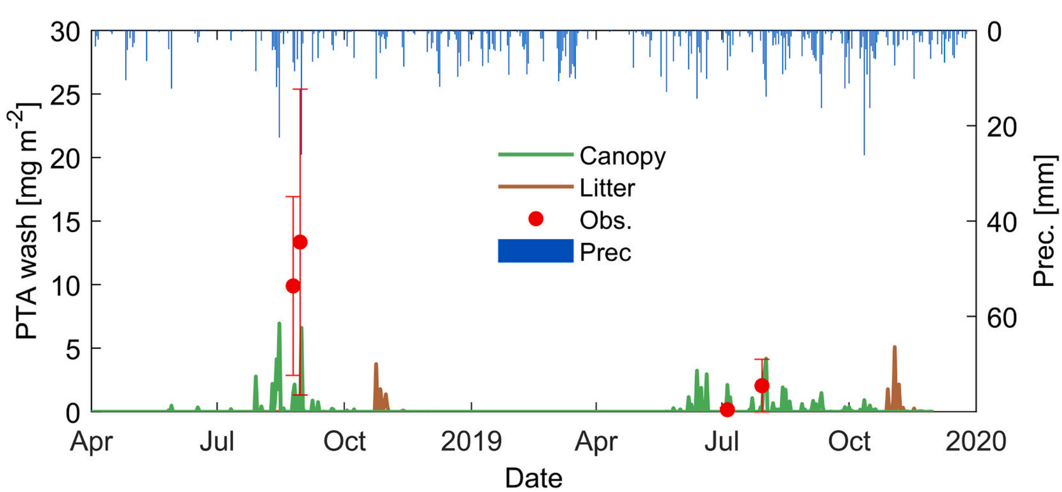

Fig. 4. Model fit of simulated PTA mass washed off during precipitation events (blue line), from both the canopy (green line) and litter (brown line). Experimental observations for PTA washed off (points) during four precipitation events (n = 9–12). Average and standard deviation indicated by the points and error bars, respectively. Observation data from García-Jorgensen et al. (2021).

reaches maturity (i.e. July), when both PTA contents and LAI are at their maximum. The maximum simulated PTA mass washed off takes place during a single precipitation event in August 2018, amounting to a total of 7 mg m^{-2}.

The simulated PTA mass washed off is comparable to the mass estimated experimentally, with values within the uncertainty for all measured precipitation events in 2019 (García-Jorgensen et al., 2021). However, there is an underestimation of the simulated wash off during measured precipitation events taking place late in the 2018 growing season, where the maximum amounts washed off were observed in the field (13 mg m^{-2}). The simulated PTA wash off from litter takes place during short period of time right after senescence, with amounts up to 5 mg m^{-2} in single rain events (Fig. 4), as calculated by Daisy.

The underestimation of PTA wash off for the precipitation events in 2018 is attributed to the exceptionally dry and hot summer that year. In reality, only a fraction of the total toxin content is assumed to be available for wash off rather than the total toxin content in the canopy. This available fraction on the surface of plants (e.g. secretions, trichomes) is expected to be highly affected by the environmental conditions of that specific year (Huchelmann et al., 2017; Tattini et al., 2000; Mayer et al., 2013). For instance, an extended period with high radiation and low precipitation might lead to high maximum toxin contents, explained by both a higher toxin production and no mass being released from the canopy during dry period. Intense precipitation events under these conditions can be expected to be the worst case-scenario in terms of toxin loads to the soil.

This variability of the available fraction of toxin on the plant surface suggests that the f_{wc} coefficient (Eq. (2)) should not be constant, but rather a function of several factors other than LAI. Some proposed factors affecting the wash off are phenology (described with DS in DAISY), water stress and rain erosivity (i.e. kinetic energy of raindrop based on drop-size). Therefore, the implementation of these factors in the wash off function (Eq. (3)) is expected to improve substantially the prediction of PTA wash off, but more data are required before such optimization can be accomplished.

3.3. Soil water

The overall simulation of soil water content offers a good description of the observations, with simulated values usually within the standard deviation of the observations (SI, Fig. S3). The high standard deviation of the observed values for the soil depth of 34 cm reflects the high heterogeneity in the water content measured with distances of 1 m (i.e. the distance between sensors). Model simulations shows that the soil characteristics in Humleore gives place to fast flow of water through macropores during intense precipitation. There was 3-fold higher cumulative volume of water percolated through matrix than by macropore flow (SI, Fig. S4), which for the 2019 hydrological year March 2019–March 2020 accounted for 182 and 52 mm, respectively (see SI, Fig. S4).

The assumption of soils with static SHPs leads to a smoothening of the model results compared with water dynamics in the field. In reality, soil structure and SHPs are dynamic and affected by wetting and drying, and this may have a significant influence on preferential flow paths (Holten et al., 2018; Vereecken et al., 2016; Šimůnek et al., 2003). Moreover, fluctuations in the groundwater table might be an explanation for the acute rise in water contents at 34 cm in the field observations towards the beginning of the year 2020 (SI, Fig. S2). The fluctuation of the groundwater in the soil is not included in the current parametrization of the soil profile Daisy, due to lack of observations.

3.4. PTA concentrations in the soil solution

Leaching of PTA takes place in form of pulses with high concentrations driven by precipitation events. This is in agreement with the fast transport of PTA observed in previous studies, both under laboratory and field conditions (Clauson-Kaas et al., 2016; Rasmussen et al., 2005). The model provides a good description of the soil water PTA concentrations, with the presence of two major pulses taking place in July and October 2019 (Fig. 5). The maximum simulated concentrations at 50 cm depth for these pulses are 883 and 919 ng L^{-1} for the events in July and October, respectively.

The pulse observed in July is explained by a heavy precipitation event taking place at the stage of maximum PTA content in the canopy (Fig. 5, dotted line). During this event, transport of PTA took place by preferential flow thus explaining the high concentrations in the leachate. The pulse in October is explained by the fast release of the remaining PTA content when the plant suffers senescence, turning into litter. Even though litter contribute to the release of PTA during a short period of time, it might contribute significantly to groundwater contamination as it takes place during more favorable leaching conditions. A plausible explanation is the occurrence of the first severe frost of the growing season, hence leading to cell lysis and a sudden release of the remaining PTA content in the litter. Moreover, cycles of freezing and thawing have been correlated to an increased leaching of pesticides, due to increased preferential transport in partially frozen soils (Holten et al.,

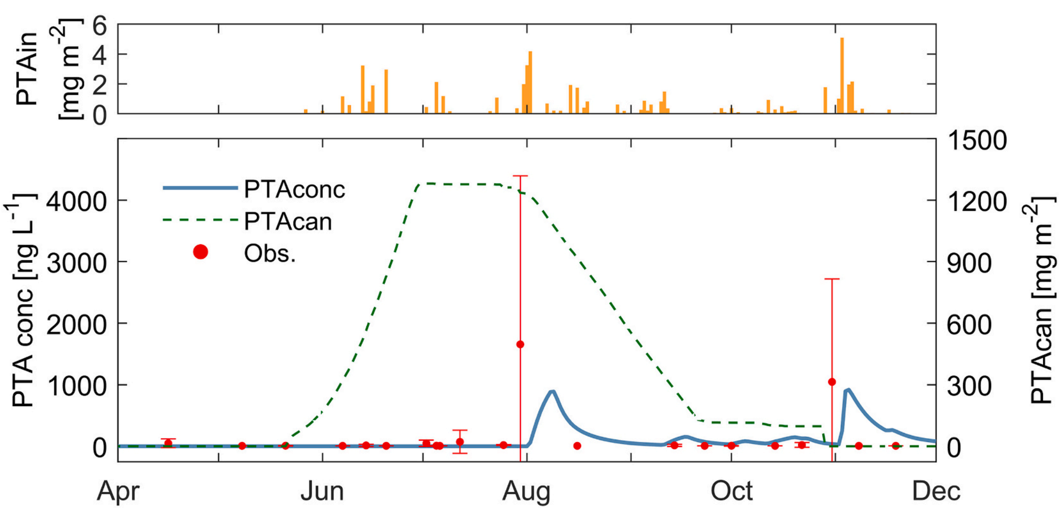

Fig. 5. Top: simulated PTA input to the soil by rain wash off, from both canopy and litter in the year 2019. Bottom: model fit of PTA concentrations in the soil at 50 cm depth (PTAconc). Simulated data (blue line) is compared to the observed concentrations (red dots) in the soil solution. Simulated PTA contents in the canopy (PTAcan) during the same period is shown as dotted green line. Data from García-Jorgensen et al. (2021).

2018). Taking into consideration a potential high release of PTA, fast transport and low degradation in the soil, the first days of frost in the autumn have the potential to generate pulses with high concentrations of PTA.

There is a slight delay of the simulated concentrations compared with the observations for both events, indicating a slightly faster transport in the field compared with the model (Fig. 5). This might be derived from the sampling of soil water with suction cells, where small pore sizes of the suction cells are only capable of sampling a narrow range of soil pore sizes in the close vicinity to the cells. Previous research showed that sampling soil solution by suction cups underestimate both the fast-flowing water (macropores) and the slow-flowing water (inter-clay pores) (Wey et al., 2022; Singh et al., 2018). These observations, together with the fact that PTA is degrading rapidly in soil solution, suggest that suction cups are not ideal sampling techniques to capture the leaching of PTA through macropores. Weighable lysimeters may offer a promising solution because the leaching mass fluxes can be estimated with high precision (Groh et al., 2020). Another uncertainty may arise from the use of hourly precipitation data, which may have underestimated macropore flow. Short intense summer showers like the ones experienced in August 2018 are events with effects on the scale of minutes. Hence, absolute results for macropore transport are highly uncertain and should be considered with caution.

3.5. Overall fate of PTA in the environment

Table 2 provides simulated data for the annual mass balance of PTA at the pedon-scale, calculated at a soil depth of 50 cm. The standard deviation for each pool or process is calculated from the differences between the years considered in the simulations (2015–2019) (SI, Fig. S5). Results show an annual PTA generation of 1299 ± 13 mg m^{-2}, with a variation explained by different biomass maximum reached in different years.

From all the PTA produced in the canopy, only 4.4 % is released by precipitation from both canopy and litter, where 4.2 % of the PTA ends up entering the soil profile. The remaining PTA is degraded on the soil surface. Please note that in reality a fraction of PTA will be degraded directly in the canopy, based on the presence of the degradation product of PTA in the canopy, pterosin B (PTB) (García-Jorgensen et al., 2021). Degradation of PTA in the canopy is set to zero and therefore, not included in the calculations.

Once in the soil, degradation is by far the most important process contributing to the fate of PTA in the soil, removing 99.7 % of the mass of PTA added to the soil. The high amounts of PTA being degraded rapidly in the soil are in agreement with previous studies (Rasmussen et al., 2003a; García-Jorgensen et al., 2021), which found negligible amounts of PTA remaining after few days in contact with soil materials. Microbial degradation and to a much lesser extent hydrolysis, are the two mechanisms that are expected to contribute to the transformation of PTA in the soil.

Table 2
Mass balance of ptaquiloside (PTA) at the pedon-scale in Humleore, showing the average annual mass, standard deviation (SD) and the percentage of each pool or process to the total. Data from simulations for the year 2015–2019 (n = 5). Values calculated as total average annual mass, with standard deviations (SD).

	PTA mg m^{-2}		%
	Aver	SD	
PTA generation	1298.5	13.2	100.0
Litter degradation	40.6	7.2	3.1
Canopy wash	48.8	6.2	3.8
Litter wash	8.3	7.3	0.6
Surface degradation	4.2	1.0	0.3
Total input soil	**54.4**	**12.6**	**4.2**
Soil degradation	54.3	12.5	99.7
Leak matrix	0.05	0.03	0.09
Leak macropore	0.12	0.07	0.22
Total leaching	**0.17**	**0.10**	**0.31**

The mass of PTA leaching through the soil profile, both by matrix and macropores, only accounts for an average of 0.013 % of the total amount of PTA produced in the canopy annually (a total of 0.17 mg PTA leached m^{-2} divided by 1298.5 mg PTA produced m^{-2}) . Transport of PTA through macropores plays an important role in the overall simulated leaching of PTA, being responsible for 72 % of the total mass of PTA leaching through the rhizosphere (0.17 mg m^{-2}). Macropore transport can thus be related to the major pulses observed in the field, considering that the fresh input of PTA to the soil suffers minimal degradation and dilution (García-Jorgensen et al., 2021). For the five years considered, maximum soil solution concentrations ranged from 1124 to 4330 ng L^{-1} (SI, Fig. S4). This concentration range for PTA indicates that pulse events are often exceeding the 16 ng L^{-1} threshold by 2 order of magnitude, hence posing a considerable carcinogenic risk towards humans if shallow groundwater is used as drinking water (Rasmussen et al., 2005).

4. Conclusion

This study presents the first calibrated model suitable for simulating the environmental fate of PTA. The mechanistic model DAISY offers a good description of the observations regarding bracken biomass and PTA generation in the canopy during the years 2018–2019. The good model fit reflects the ability of the PTA generation function to explain toxin contents based on biomass and development stage alone. The results support the rationale of using a mechanistic model to assess the environmental fate of natural pollutants compared to other more simplistic approaches which do not simulate plant growth. Maximum potential wash off events are identified as intense precipitation events taking place in the middle of summer, at the time where maximum plant biomass, LAI and phytotoxin content has been reached. Degradation in the soil is the most important process determining fate of PTA in the soil, removing over 99 % of the total PTA input to the soil.

Leaching of PTA, under the conditions considered in this study, is concentrated during the summer period and beginning of autumn, taking place through preferential transport in form of pulses with simulated concentrations up to 4390 ng L^{-1}. Transport through macropores was found to be responsible for 72 % of the PTA leached to groundwater, which is explained by the fast transport and thus minimal time for dilution and degradation.

Simulated values show that litter, even though it presents only a small fraction of the maximum PTA content recorded in the season, has the potential to produce PTA leaching at high concentrations in the beginning of the autumn and this process requires further investigation. The higher water content in the soil profile, lower temperatures and shorter residence time in the soil, makes the beginning of autumn an ideal period for leaching as it has been observed for pesticides (Rosenbom et al., 2015). Based on the results it is estimated that, on average, only 0.009 % of the total amount of PTA produced in the canopy leaches to the groundwater. However, this small fraction of PTA may result in relatively high water concentrations in terms of toxicity.

More research is needed to improve the description of toxin release from plants, calling for specific attention towards the fraction of toxin available for wash off on the plant surface, rather than the total plant toxin content. Moreover, scaling factors on the generation functions should be included for stress, development stage, precipitation intensity and cumulative precipitation.

CRediT authorship contribution statement

Daniel B. García-Jorgensen: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Validation, Visualization, Writing – original draft, Writing – review & editing. **Maja Holbak:** Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Writing – review & editing. **Hans Christian Bruun**

Hansen: Conceptualization, Funding acquisition, Methodology, Project administration, Resources, Supervision, Writing – review & editing. **Per Abrahamsen:** Conceptualization, Resources, Software. **Efstathios Diamantopoulos:** Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Supervision, Validation, Visualization, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.scitotenv.2024.170658.

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